

Hello everyone, today we are going to present our project called Library Management System.
This project was developed as part of our ITP course.
Our goal was to create a simple and efficient system to manage books and users in a library.

Managing a library manually can be difficult and inefficient. It is hard to track which books are available and which are borrowed.

Also, there is no easy way to store user activity and history.

```
from models import Book, User
from storage import load_json, save_json
class Library: 2+ usages  ⌘ Akylsolo +1
    def __init__(self, books_file="books.json", users_file="users.json"):  ⌘ Akylsolo
        self.books_file = books_file
        self.users_file = users_file
        self.books = self.load_books()
        self.users = self.load_users()
    def load_books(self): 1 usage  ⌘ Akylsolo
        data = load_json(self.books_file)
        return {
            book["id"]: Book(
                book["id"],
                book["title"],
                book["author"],
                book["available"]
            )
            for book in data
        }
    def load_users(self): 1 usage  ⌘ Akylsolo
        data = load_json(self.users_file)
        return {
            user["user_id"]: User(
                user["user_id"],
                user["name"],
                user.get("borrowed_books", []),
                user.get("history", [])
            )
            for user in data
        }
```


To solve this problem, we created a console-based application in Python.

Our system allows users to manage books, track borrowing and returning, and store data using JSON files.

This makes the process faster, easier, and more organized.



```
from library import Library
def show_menu(): 1 usage  3jl
    print("\nLibrary Management System")
    print("1. Show all books")
    print("2. Show available books")
    print("3. Show borrowed books")
    print("4. Search book")
    print("5. Borrow book")
    print("6. Return book")
    print("7. Show user history")
    print("8. Show statistics")
    print("9. Exit")
def print_books(books :{__iter__}): 4+ usages  3jl
    if not books:
        print("No books found")
        return
    for book in books:
        print(book.get_info())
def main(): 1 usage  3jl
    library = Library()
    while True:
        show_menu()
        choice = input("Choose option: ")
        try:
            if choice == "1":
                print_books(library.book_generator())
            elif choice == "2":
                print_books(library.get_available_books())
            elif choice == "3":
```




Our system includes several important features:

- Users can view all books and available books.

- They can borrow and return books.

- They can search books by title or author.

- The system also tracks user history and shows statistics.


```
out << "Enter rows and columns for second matrix: ";
cin >> r2 >> c2;

// Taking elements of first matrix.
out << "Enter elements of matrix 1:" << endl;
for(i = 0; i < r1; ++i)
    for(j = 0; j < c1; ++j)
    {
        out << "Enter element a" << i + 1 << j + 1 << " : ";
        cin >> a[i][j];
    }

// Taking elements of second matrix.
out << "Enter elements of matrix 2:" << endl;
for(i = 0; i < r2; ++i)
    for(j = 0; j < c2; ++j)
    {
        out << "Enter element b" << i + 1 << j + 1 << " : ";
        cin >> b[i][j];
    }
```

We used Python as the main programming language.

We applied Object-Oriented Programming concepts.

We used JSON files for data storage.

We also used dictionaries and lists for data management.

Additionally, we implemented a generator and a decorator.

Our system is built using several classes:

LibraryItem, Book, DigitalBook, User, and Library.

Each class has its own responsibility, which makes the code clean and organized.

```
resp_iter = self.stub.GetDatacenterStatuses()
statuses = {}
for data in resp_iter:
    status = Status(
        status_id=data.id, name=data.name
    )
    statuses[status.name] = status
return statuses
```




We used key OOP concepts in our project.

Inheritance: Book inherits from LibraryItem, and DigitalBook inherits from Book.

Polymorphism: the method `get_info()` works differently in each class.

Encapsulation: we used private attributes and controlled access to da

We used a dictionary to store books and users because it allows fast access by ID.

We used lists to store borrowed books and user history.

We used JSON files to save data permanently.





When the program starts, it loads data from JSON files.

The user interacts with the system through a menu.



The user can choose actions like borrowing or returning books.

All changes are saved automatically.

```
Library Management System
1. Show all books
2. Show available books
3. Show borrowed books
4. Search book
5. Borrow book
6. Return book
7. Show user history
8. Show statistics
9. Exit
Choose option: 1
1. Harry Potter by J.K.Rowling - Available
2. The Lord of the Rings by J.R.R. Tolkien - Available
3. The Good Witch Azura by Mildred Featherwhyte - Available
4. The Hunger Games by Suzanne Collins - Available
```

```
Library Management System
1. Show all books
2. Show available books
3. Show borrowed books
4. Search book
5. Borrow book
6. Return book
```




In conclusion, our Library Management System is simple, efficient, and easy to use.

It demonstrates the use of Python, OOP, and data management. In the future, it can be improved by adding a graphical interface or database.

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Thank you